

LAPORAN SQL ANALYSIS

E-Commerce Sales Dataset — Day 3

Roadmap 7 Hari Data Analyst Portfolio — Dokumentasi Lengkap Langkah demi Langkah

By : Alya Shandy

1. Pendahuluan

Laporan ini mendokumentasikan proses analisis data pada dataset e-commerce sales yang sudah dibersihkan di Day 1 (cleaned_sales_data.csv, 3.933 baris). Analisis dilakukan sepenuhnya menggunakan SQL (MySQL Server & MySQL Workbench), sesuai checklist Day 3 dalam roadmap Data Analyst portfolio 7 hari.

Sama seperti laporan Day 2, laporan ini disusun sebagai tutorial langkah demi langkah — setiap bagian menjelaskan CARA melakukan sesuatu di MySQL Workbench (menu apa yang diklik, opsi wizard apa yang dicentang, query apa yang ditulis), bukan cuma menampilkan hasil akhirnya. Tujuannya supaya proses ini bisa diulang lagi atau diajarkan ke orang lain tanpa harus mengingat dari nol.

File kerja: business_analysis.sql — berisi 15 query bisnis yang dijalankan terhadap tabel sales_data di database ecommerce_sales, hasil import dari cleaned_sales_data.csv.

2. Ringkasan Eksekutif

Tabel 1 merangkum KPI utama hasil analisis SQL.

Metrik	Nilai
Total Revenue (Completed)	\$252.227,71
Total Orders (Completed)	3.220
Average Order Value	\$78,33
Average Revenue per Customer	\$288,59
Kontribusi Region East	74,4% dari total revenue
Kontribusi Kategori Electronics	25,33% dari total revenue

Tabel 1. KPI utama Sales Performance Summary

3. Tahapan Analisis (Tutorial)

3.1 Instalasi MySQL Server & Workbench

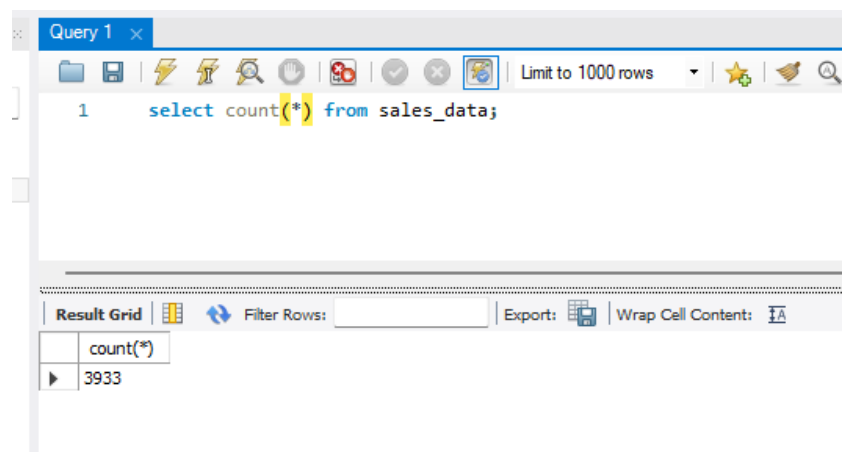
Sebelum bisa menjalankan query, MySQL Server dan MySQL Workbench perlu terpasang terlebih dahulu.

1. Unduh MySQL Installer for Windows dari situs resmi MySQL, pilih versi lengkap (sekitar 565.9 MB) yang sudah menyertakan MySQL Server dan MySQL Workbench sekaligus.

2. Jalankan installer, lalu pada pilihan setup type pilih "Full" agar semua komponen (server, workbench, connector) terpasang sekaligus.
3. Lanjutkan proses instalasi mengikuti wizard bawaan — opsi tambahan yang tidak diperlukan tidak perlu dicentang.
4. Buat root password untuk MySQL Server pada langkah konfigurasi akun. Password ini dipakai setiap kali connect ke database.
5. Selesaikan instalasi hingga klik "Finish", lalu buka MySQL Workbench dan buat koneksi baru menggunakan password root yang sudah dibuat.

3.2 Membuat Tabel & Import Data

1. Buat database baru (schema) bernama ecommerce_sales, lalu buat tabel sales_data dengan kolom sesuai struktur dataset.
2. Klik kanan tabel sales_data di panel kiri → pilih "Table Data Import Wizard".
3. Klik "Browse...", pilih file cleaned_sales_data.csv, lalu klik Next.
4. Pastikan "Use existing table" sudah otomatis terpilih ke ecommerce_sales.sales_data, lalu centang "Truncate table before import" supaya tabel dikosongkan dulu sebelum diisi — berguna untuk mencegah data dobel bila import diulang.
5. Klik Next lalu Finish untuk menjalankan import, lalu verifikasi jumlah baris yang masuk.



Gambar 1. Verifikasi jumlah baris data ter-import — 3.933 baris, sama persis dengan hasil cleaning Day 1

Kolom order_date bertipe varchar (bukan DATE), sehingga setiap kali butuh analisis berbasis waktu (bulanan, harian), kolom ini perlu dikonversi dulu dengan fungsi STR_TO_DATE().

Limit to 1000 rows

```

1 describe sales_data;
2

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Field	Type	Null	Key	Default	Extra
▶	order_id	varchar(15)	YES		NULL	
	order_date	varchar(10)	YES		NULL	
	customer_id	varchar(15)	YES		NULL	
	customer_name	varchar(100)	YES		NULL	
	region	varchar(20)	YES		NULL	
	category	varchar(30)	YES		NULL	
	product_name	varchar(50)	YES		NULL	
	quantity	int	YES		NULL	
	unit_price	decimal(10,2)	YES		NULL	
	order_status	varchar(20)	YES		NULL	
	payment_method	varchar(20)	YES		NULL	
	revenue	decimal(10,2)	YES		NULL	

Gambar 2. Struktur tabel (DESCRIBE sales_data) — 12 kolom dengan tipe data masing-masing

```

1 select * from sales_data limit 5;
2

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#) | Fetch rows: [10](#)

	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	revenue
▶	ORD100363	6/17/2023	CUST1454	Fajar Wijaya	East	Beauty	Lip Balm	1	6.00	Completed	Gift Card	6.00
	ORD101186	10/17/2024	CUST1674	Dewi Lee	East	Beauty	Lip Balm	1	6.22	Completed	Gift Card	6.22
	ORD101364	7/11/2024	CUST1201	Linda Johnson	East	Beauty	Shampoo	2	6.22	Completed	Credit Card	12.44
	ORD102845	11/20/2023	CUST1361	Rina Brown	East	Beauty	Moisturizer	2	6.27	Completed	Debit Card	12.54
	ORD102156	1/11/2024	CUST1798	Siti Anderson	East	Beauty	Lip Balm	1	6.38	Completed	Gift Card	6.38

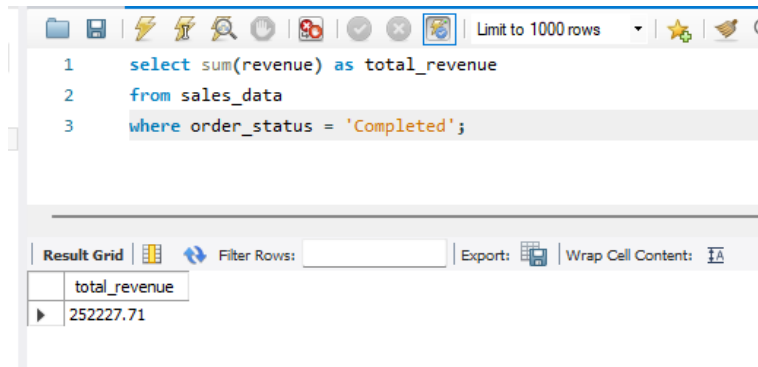
Gambar 3. Contoh 5 baris data (SELECT * LIMIT 5) — struktur kolom sesuai hasil cleaning Day 1

3.3 Query 1 — Total Revenue

```

SELECT SUM(revenue) AS total_revenue
FROM sales_data
WHERE order_status = 'Completed';

```



Gambar 4. Hasil Query 1 — total_revenue = 252.227,71

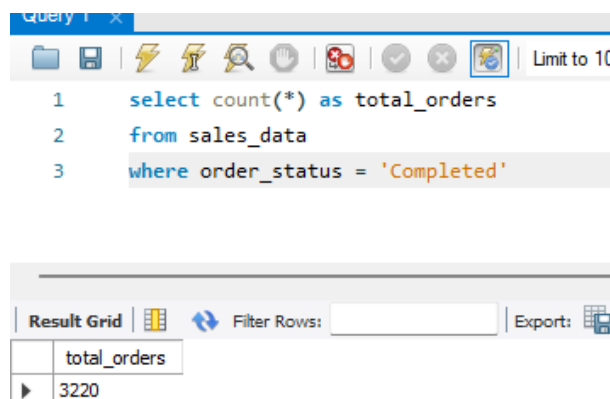
Filter WHERE order_status = 'Completed' dipakai karena order yang di-cancel/return tidak dihitung sebagai revenue riil bisnis.

3.4 Query 2 — Total Orders

```

SELECT COUNT(*) AS total_orders
FROM sales_data
WHERE order_status = 'Completed';

```



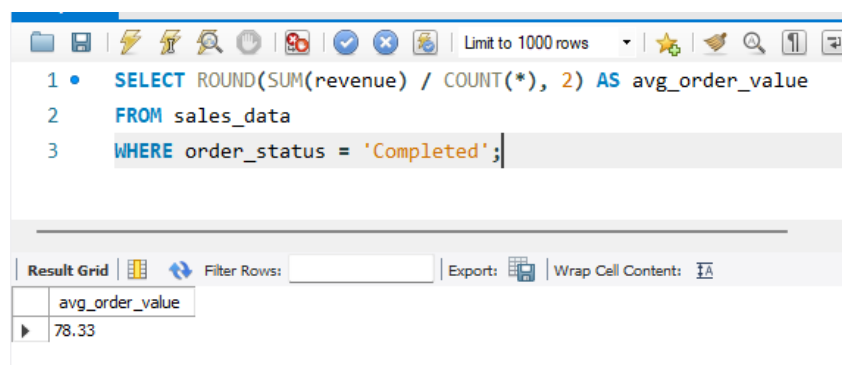
Gambar 5. Hasil Query 2 — total_orders = 3.220

3.5 Query 3 — Average Order Value (AOV)

```

SELECT ROUND(SUM(revenue) / COUNT(*), 2) AS avg_order_value
FROM sales_data
WHERE order_status = 'Completed';

```



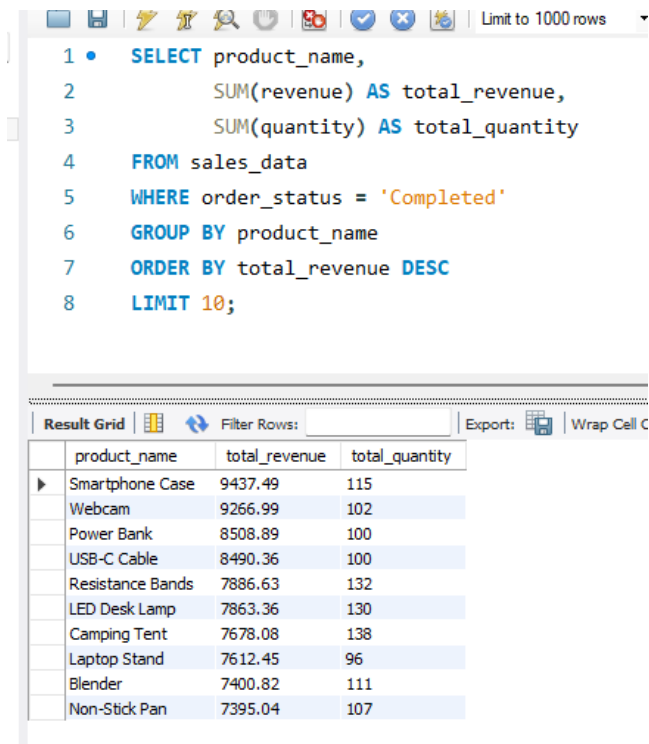
Gambar 6. Hasil Query 3 — avg_order_value = 78,33

3.6 Query 4 — Top 10 Products by Revenue

```

SELECT product_name,
       SUM(revenue) AS total_revenue,
       SUM(quantity) AS total_quantity
FROM sales_data
WHERE order_status = 'Completed'
GROUP BY product_name
ORDER BY total_revenue DESC
LIMIT 10;

```



The screenshot shows a SQL query editor with a toolbar at the top. The query is entered in the main text area. Below the query, a 'Result Grid' is displayed, showing the results of the query. The table has four columns: product_name, total_revenue, and total_quantity. The results are sorted by total_revenue in descending order, with 'Smartphone Case' at the top.

product_name	total_revenue	total_quantity
Smartphone Case	9437.49	115
Webcam	9266.99	102
Power Bank	8508.89	100
USB-C Cable	8490.36	100
Resistance Bands	7886.63	132
LED Desk Lamp	7863.36	130
Camping Tent	7678.08	138
Laptop Stand	7612.45	96
Blender	7400.82	111
Non-Stick Pan	7395.04	107

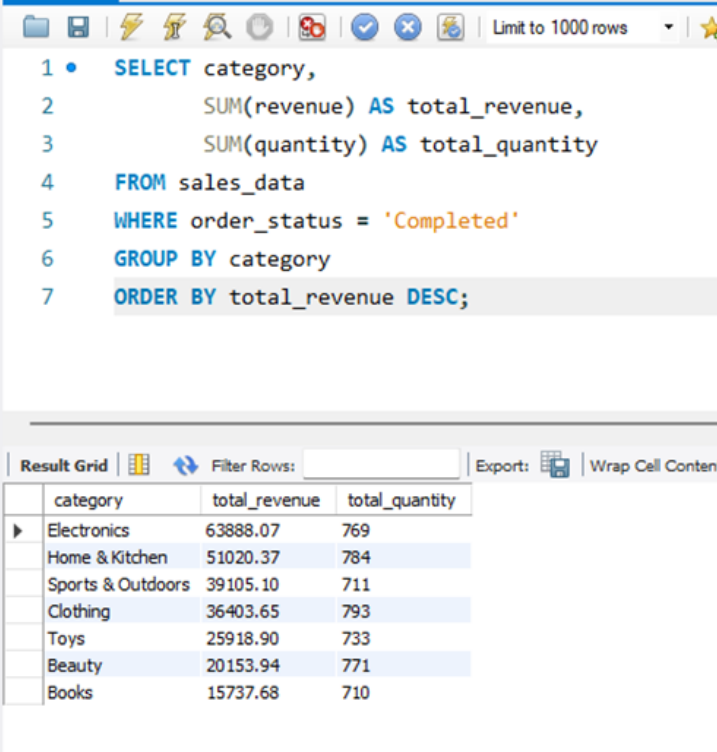
Gambar 7. Hasil Query 4 — Smartphone Case memimpin dengan revenue 9.437,49

3.7 Query 5 — Top Categories

```

SELECT category,
       SUM(revenue) AS total_revenue,
       SUM(quantity) AS total_quantity
FROM sales_data
WHERE order_status = 'Completed'
GROUP BY category
ORDER BY total_revenue DESC;

```



```

1 • SELECT category,
2       SUM(revenue) AS total_revenue,
3       SUM(quantity) AS total_quantity
4 FROM sales_data
5 WHERE order_status = 'Completed'
6 GROUP BY category
7 ORDER BY total_revenue DESC;

```

category	total_revenue	total_quantity
Electronics	63888.07	769
Home & Kitchen	51020.37	784
Sports & Outdoors	39105.10	711
Clothing	36403.65	793
Toys	25918.90	733
Beauty	20153.94	771
Books	15737.68	710

Gambar 8. Hasil Query 5 — Electronics memimpin dengan revenue 63.888,07

3.8 Query 6 — Monthly Sales

```

SELECT DATE_FORMAT(STR_TO_DATE(order_date, '%m/%d/%Y'), '%Y-%m') AS month,
       SUM(revenue) AS total_revenue,
       COUNT(*) AS total_orders
FROM sales_data
WHERE order_status = 'Completed'
GROUP BY month
ORDER BY month;

```

Query 1

```

1 • SELECT DATE_FORMAT(STR_TO_DATE(order_date, '%m/%d/%Y'), '%Y-%m') AS month,
2       SUM(revenue) AS total_revenue,
3       COUNT(*) AS total_orders
4 FROM sales_data
5 WHERE order_status = 'Completed'
6 GROUP BY month
7 ORDER BY month;

```

Result Grid

month	total_revenue	total_orders
2023-01	10999.53	144
2023-02	11485.76	131
2023-03	13600.22	150
2023-04	10068.15	142
2023-05	12690.06	138
2023-06	7248.67	103
2023-07	11168.56	133
2023-08	11470.69	135
2023-09	12740.75	144
2023-10	11436.65	138
2023-11	9873.17	129
2023-12	7075.01	106
2024-01	9416.06	119
2024-02	8862.26	122
2024-03	10659.45	137
2024-04	9007.92	133
2024-05	12166.04	151
2024-06	8924.90	114
2024-07	9983.49	134
2024-08	10466.35	139
2024-09	8546.15	130
2024-10	11485.51	157
2024-11	11660.19	147
2024-12	11192.17	144

Gambar 9. Hasil Query 6 — revenue bulanan Jan 2023 s.d. Des 2024 (24 bulan)

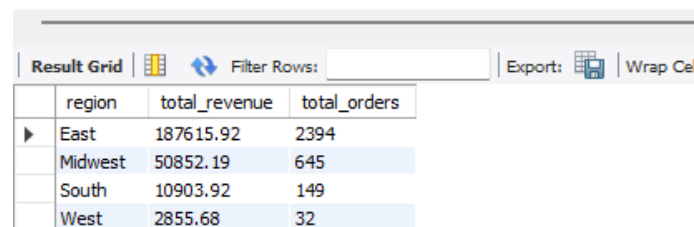
3.9 Query 7 — Regional Sales

```

SELECT region,
       SUM(revenue) AS total_revenue,
       COUNT(*) AS total_orders
FROM sales_data
WHERE order_status = 'Completed'
GROUP BY region
ORDER BY total_revenue DESC;

```

```
1 • SELECT region,  
2       SUM(revenue) AS total_revenue,  
3       COUNT(*) AS total_orders  
4 FROM sales_data  
5 WHERE order_status = 'Completed'  
6 GROUP BY region  
7 ORDER BY total_revenue DESC;
```

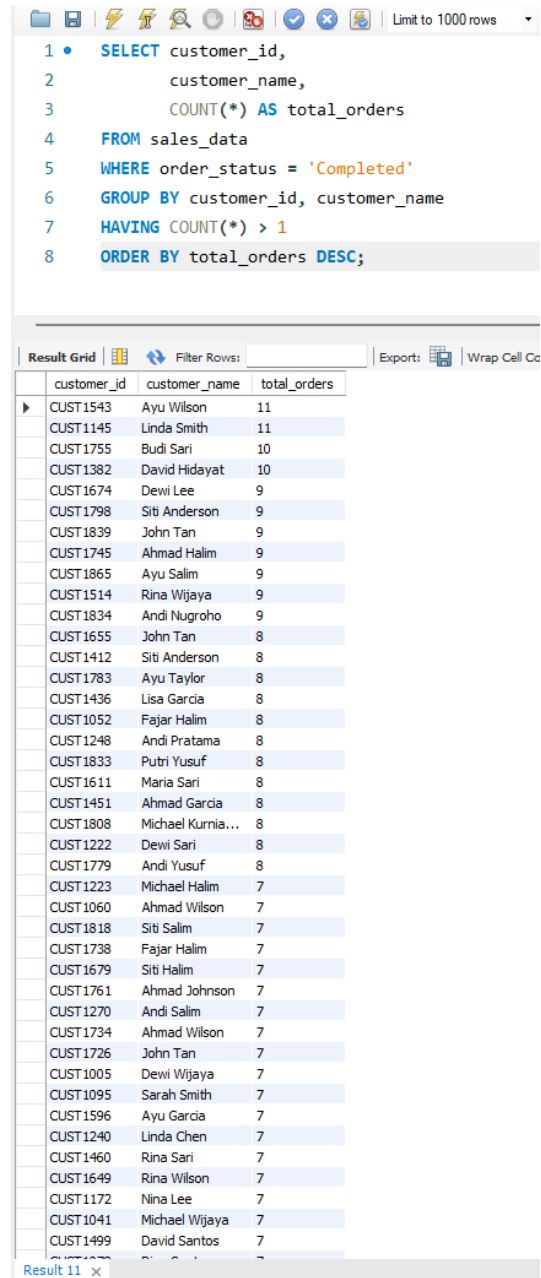


	region	total_revenue	total_orders
►	East	187615.92	2394
	Midwest	50852.19	645
	South	10903.92	149
	West	2855.68	32

Gambar 10. Hasil Query 7 — Region East mendominasi (187.615,92 dari total 252.227,71)

3.10 Query 8 — Repeat Customers

```
SELECT customer_id,
       customer_name,
       COUNT(*) AS total_orders
FROM sales_data
WHERE order_status = 'Completed'
GROUP BY customer_id, customer_name
HAVING COUNT(*) > 1
ORDER BY total_orders DESC;
```



1 • SELECT customer_id,
2 customer_name,
3 COUNT(*) AS total_orders
4 FROM sales_data
5 WHERE order_status = 'Completed'
6 GROUP BY customer_id, customer_name
7 HAVING COUNT(*) > 1
8 ORDER BY total_orders DESC;

Result Grid | Filter Rows: | Exports: | Wrap Cell Co

customer_id	customer_name	total_orders
CUST1543	Ayu Wilson	11
CUST1145	Linda Smith	11
CUST1755	Budi Sari	10
CUST1382	David Hidayat	10
CUST1674	Dewi Lee	9
CUST1798	Siti Anderson	9
CUST1839	John Tan	9
CUST1745	Ahmad Halim	9
CUST1865	Ayu Salim	9
CUST1514	Rina Wijaya	9
CUST1834	Andi Nugroho	9
CUST1655	John Tan	8
CUST1412	Siti Anderson	8
CUST1783	Ayu Taylor	8
CUST1436	Lisa Garcia	8
CUST1052	Fajar Halim	8
CUST1248	Andi Pratama	8
CUST1833	Putri Yusuf	8
CUST1611	Maria Sari	8
CUST1451	Ahmad Garcia	8
CUST1808	Michael Kurnia...	8
CUST1222	Dewi Sari	8
CUST1779	Andi Yusuf	8
CUST1223	Michael Halim	7
CUST1060	Ahmad Wilson	7
CUST1818	Siti Salim	7
CUST1738	Fajar Halim	7
CUST1679	Siti Halim	7
CUST1761	Ahmad Johnson	7
CUST1270	Andi Salim	7
CUST1734	Ahmad Wilson	7
CUST1726	John Tan	7
CUST1005	Dewi Wijaya	7
CUST1095	Sarah Smith	7
CUST1596	Ayu Garcia	7
CUST1240	Linda Chen	7
CUST1460	Rina Sari	7
CUST1649	Rina Wilson	7
CUST1172	Nina Lee	7
CUST1041	Michael Wijaya	7
CUST1499	David Santos	7

Result 11 x

Gambar 11. Hasil Query 8 — Ayu Wilson dan Linda Smith paling sering order (11x)

3.11 Query 9 — Highest-Value Customers

```
SELECT customer_id,
       customer_name,
       SUM(revenue) AS total_spent,
       COUNT(*) AS total_orders
FROM sales_data
```

```

WHERE order_status = 'Completed'
GROUP BY customer_id, customer_name
ORDER BY total_spent DESC
LIMIT 10;

```

Query 1 x

Limit to 1000 rows

```

1 • SELECT customer_id,
2         customer_name,
3         SUM(revenue) AS total_spent,
4         COUNT(*) AS total_orders
5 FROM sales_data
6 WHERE order_status = 'Completed'
7 GROUP BY customer_id, customer_name
8 ORDER BY total_spent DESC
9 LIMIT 10;

```

Result Grid | Filter Rows: | Export: | Wrap Cell C

	customer_id	customer_name	total_spent	total_orders
▶	CUST1644	Andi Salim	1285.26	6
	CUST1054	Putri Garcia	1054.93	3
	CUST1543	Ayu Wilson	972.76	11
	CUST1755	Budi Sari	966.12	10
	CUST1248	Andi Pratama	951.12	8
	CUST1465	Nina Wilson	944.29	4
	CUST1819	Lisa Salim	938.90	4
	CUST1579	Kevin Tan	936.35	6
	CUST1870	Dewi Chen	906.64	6
	CUST1836	Kevin Salim	904.89	5

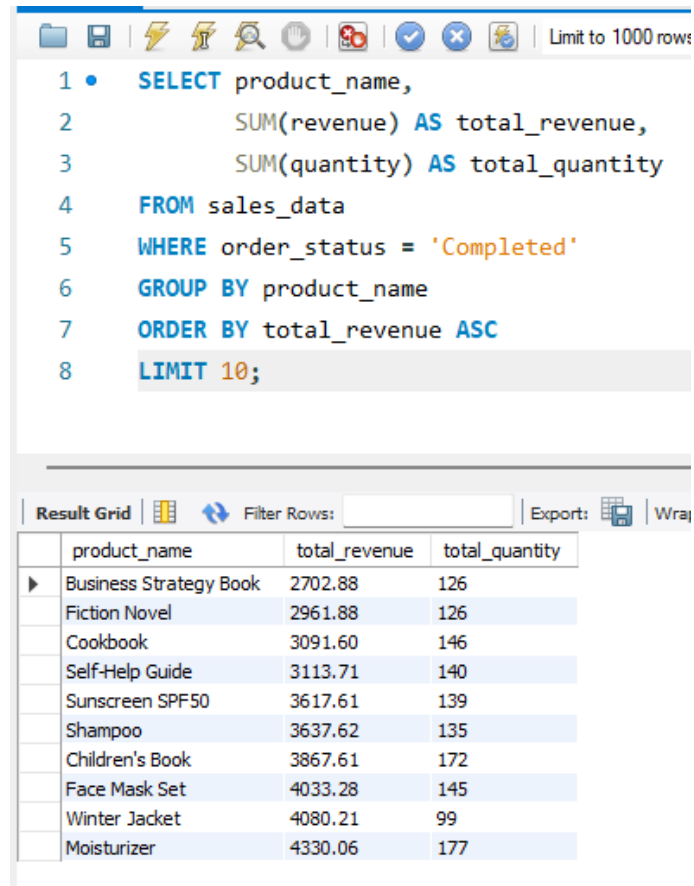
Gambar 12. Hasil Query 9 — Andi Salim total belanja tertinggi (1.285,26) meski hanya 6 order

3.12 Query 10 — Lowest-Performing Products

```

SELECT product_name,
       SUM(revenue) AS total_revenue,
       SUM(quantity) AS total_quantity
FROM sales_data
WHERE order_status = 'Completed'
GROUP BY product_name
ORDER BY total_revenue ASC
LIMIT 10;

```



```

1 • SELECT product_name,
2       SUM(revenue) AS total_revenue,
3       SUM(quantity) AS total_quantity
4 FROM sales_data
5 WHERE order_status = 'Completed'
6 GROUP BY product_name
7 ORDER BY total_revenue ASC
8 LIMIT 10;

```

product_name	total_revenue	total_quantity
Business Strategy Book	2702.88	126
Fiction Novel	2961.88	126
Cookbook	3091.60	146
Self-Help Guide	3113.71	140
Sunscreen SPF50	3617.61	139
Shampoo	3637.62	135
Children's Book	3867.61	172
Face Mask Set	4033.28	145
Winter Jacket	4080.21	99
Moisturizer	4330.06	177

Gambar 13. Hasil Query 10 — produk kategori Books mendominasi daftar revenue terendah

3.13 Query 11 — Growth Month-over-Month (MoM)

```

SELECT month, total_revenue,
       ROUND(
         (total_revenue - LAG(total_revenue) OVER (ORDER BY month))
         / LAG(total_revenue) OVER (ORDER BY month) * 100, 2
       ) AS growth_pct
FROM (
  SELECT DATE_FORMAT(STR_TO_DATE(order_date, '%m/%d/%Y'), '%Y-%m') AS month,
         SUM(revenue) AS total_revenue
  FROM sales_data WHERE order_status = 'Completed' GROUP BY month
) AS monthly ORDER BY month;

```

```

1 • SELECT month,
2     total_revenue,
3     ROUND(
4         (total_revenue - LAG(total_revenue) OVER (ORDER BY month))
5         / LAG(total_revenue) OVER (ORDER BY month) * 100, 2
6     ) AS growth_pct
7 FROM (
8     SELECT DATE_FORMAT(STR_TO_DATE(order_date, '%m/%d/%Y'), '%Y-%m') AS month,
9         SUM(revenue) AS total_revenue
10    FROM sales_data
11   WHERE order_status = 'Completed'
12   GROUP BY month
13 ) AS monthly
14 ORDER BY month;

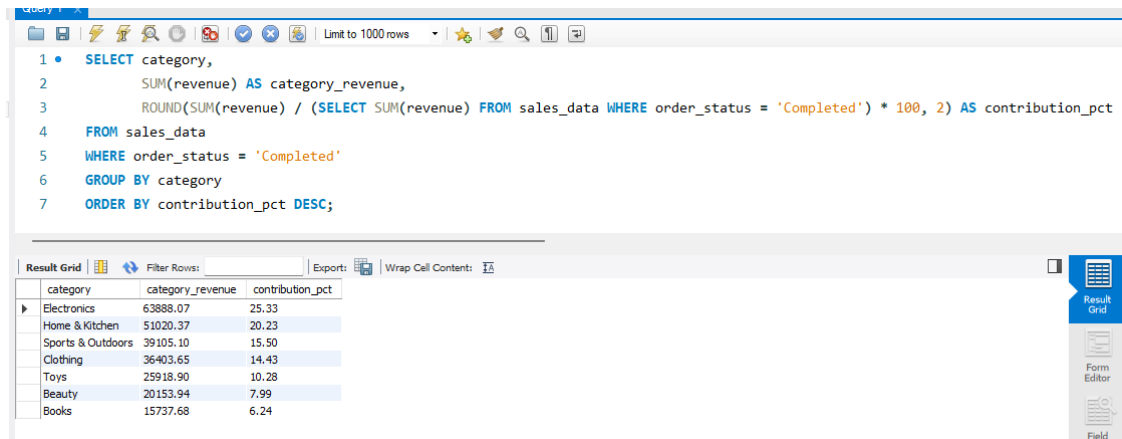
```

month	total_revenue	growth_pct
2023-01	10999.53	NULL
2023-02	11485.76	4.42
2023-03	13600.22	18.41
2023-04	10068.15	-25.97
2023-05	12690.06	26.04
2023-06	7248.67	-42.88
2023-07	11168.56	54.08
2023-08	11470.69	2.71
2023-09	12740.75	11.07
2023-10	11436.65	-10.24
2023-11	9873.17	-13.67
2023-12	7075.01	-28.34
2024-01	9416.06	33.09
2024-02	8862.26	-5.88
2024-03	10659.45	20.28
2024-04	9007.92	-15.49
2024-05	12166.04	35.06
2024-06	8924.90	-26.64
2024-07	9983.49	11.86
2024-08	10466.35	4.84
2024-09	8546.15	-18.35
2024-10	11485.51	34.39
2024-11	11660.19	1.52
2024-12	11192.17	-4.01

Gambar 14. Hasil Query 11 — window function LAG() menghitung growth bulanan (-42,88% s.d. +54,08%)

3.14 Query 12 — Category Contribution (%)

```
SELECT category, SUM(revenue) AS category_revenue,
       ROUND(SUM(revenue) / (SELECT SUM(revenue) FROM sales_data
                             WHERE order_status = 'Completed') * 100, 2) AS contribution_pct
FROM sales_data WHERE order_status = 'Completed'
GROUP BY category ORDER BY contribution_pct DESC;
```



The screenshot shows a SQL query editor with the following code:

```
1 • SELECT category,
2     SUM(revenue) AS category_revenue,
3     ROUND(SUM(revenue) / (SELECT SUM(revenue) FROM sales_data WHERE order_status = 'Completed') * 100, 2) AS contribution_pct
4 FROM sales_data
5 WHERE order_status = 'Completed'
6 GROUP BY category
7 ORDER BY contribution_pct DESC;
```

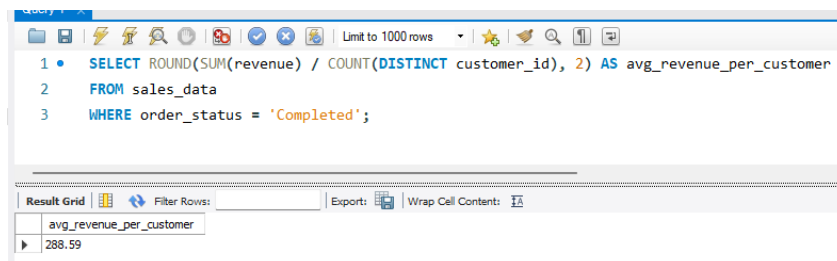
Below the code, the 'Result Grid' shows the following data:

category	category_revenue	contribution_pct
Electronics	63888.07	25.33
Home & Kitchen	51020.37	20.23
Sports & Outdoors	39105.10	15.50
Clothing	36403.65	14.43
Toys	25918.90	10.28
Beauty	20153.94	7.99
Books	15737.68	6.24

Gambar 15. Hasil Query 12 — Electronics berkontribusi 25,33% dari total revenue

3.15 Query 13 — Average Revenue per Customer

```
SELECT ROUND(SUM(revenue) / COUNT(DISTINCT customer_id), 2) AS avg_revenue_per_customer
FROM sales_data WHERE order_status = 'Completed';
```



The screenshot shows a SQL query editor with the following code:

```
1 • SELECT ROUND(SUM(revenue) / COUNT(DISTINCT customer_id), 2) AS avg_revenue_per_customer
2 FROM sales_data
3 WHERE order_status = 'Completed';
```

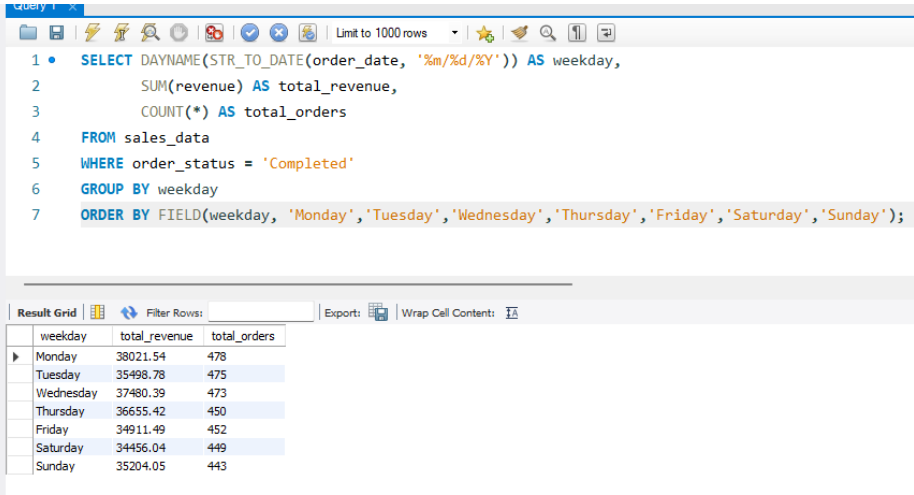
Below the code, the 'Result Grid' shows the following data:

avg_revenue_per_customer
288.59

Gambar 16. Hasil Query 13 — avg_revenue_per_customer = 288,59

3.16 Query 14 — Sales by Weekday

```
SELECT DAYNAME(STR_TO_DATE(order_date, '%m/%d/%Y')) AS weekday,
       SUM(revenue) AS total_revenue, COUNT(*) AS total_orders
FROM sales_data WHERE order_status = 'Completed'
GROUP BY weekday
ORDER BY
FIELD(weekday, 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday');
```



```

1 • SELECT DAYNAME(STR_TO_DATE(order_date, '%m/%d/%Y')) AS weekday,
2     SUM(revenue) AS total_revenue,
3     COUNT(*) AS total_orders
4 FROM sales_data
5 WHERE order_status = 'Completed'
6 GROUP BY weekday
7 ORDER BY FIELD(weekday, 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday');

```

weekday	total_revenue	total_orders
Monday	38021.54	478
Tuesday	35498.78	475
Wednesday	37480.39	473
Thursday	36655.42	450
Friday	34911.49	452
Saturday	34456.04	449
Sunday	35204.05	443

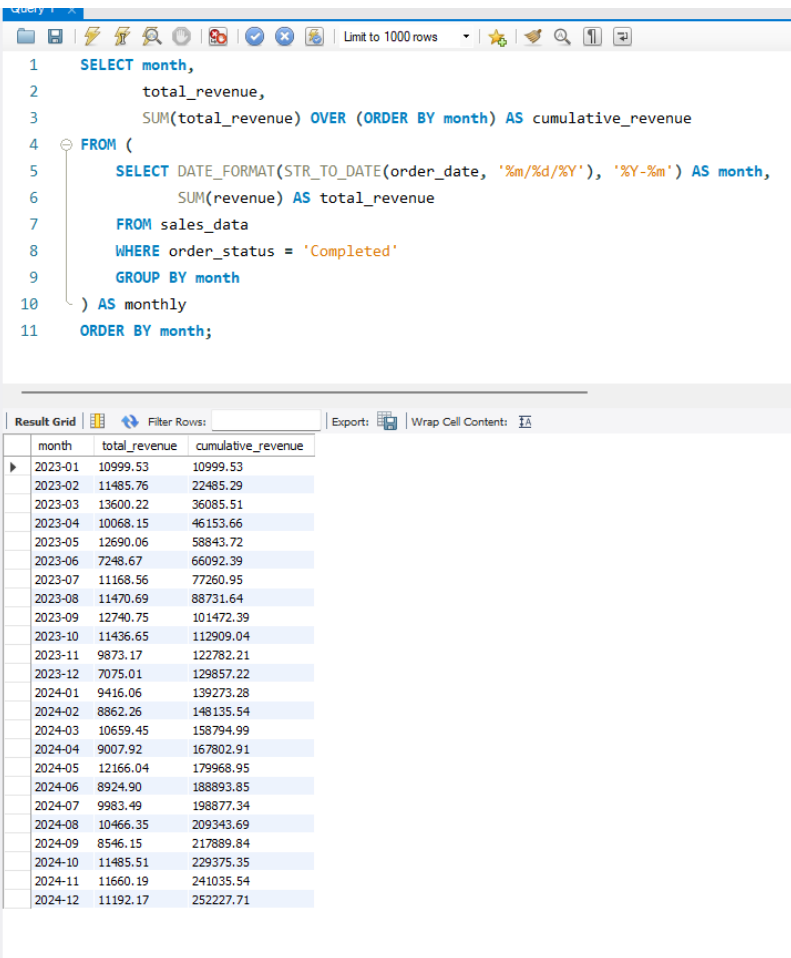
Gambar 17. Hasil Query 14 — penjualan relatif merata, Senin tertinggi dan Sabtu terendah

3.17 Query 15 — Sales Trend (Cumulative Revenue)

```

SELECT month, total_revenue,
       SUM(total_revenue) OVER (ORDER BY month) AS cumulative_revenue
FROM (
  SELECT DATE_FORMAT(STR_TO_DATE(order_date, '%m/%d/%Y'), '%Y-%m') AS month,
         SUM(revenue) AS total_revenue
  FROM sales_data WHERE order_status = 'Completed' GROUP BY month
) AS monthly ORDER BY month;

```



```

1 SELECT month,
2     total_revenue,
3     SUM(total_revenue) OVER (ORDER BY month) AS cumulative_revenue
4 FROM (
5     SELECT DATE_FORMAT(STR_TO_DATE(order_date, '%m/%d/%Y'), '%Y-%m') AS month,
6           SUM(revenue) AS total_revenue
7     FROM sales_data
8     WHERE order_status = 'Completed'
9     GROUP BY month
10 ) AS monthly
11 ORDER BY month;

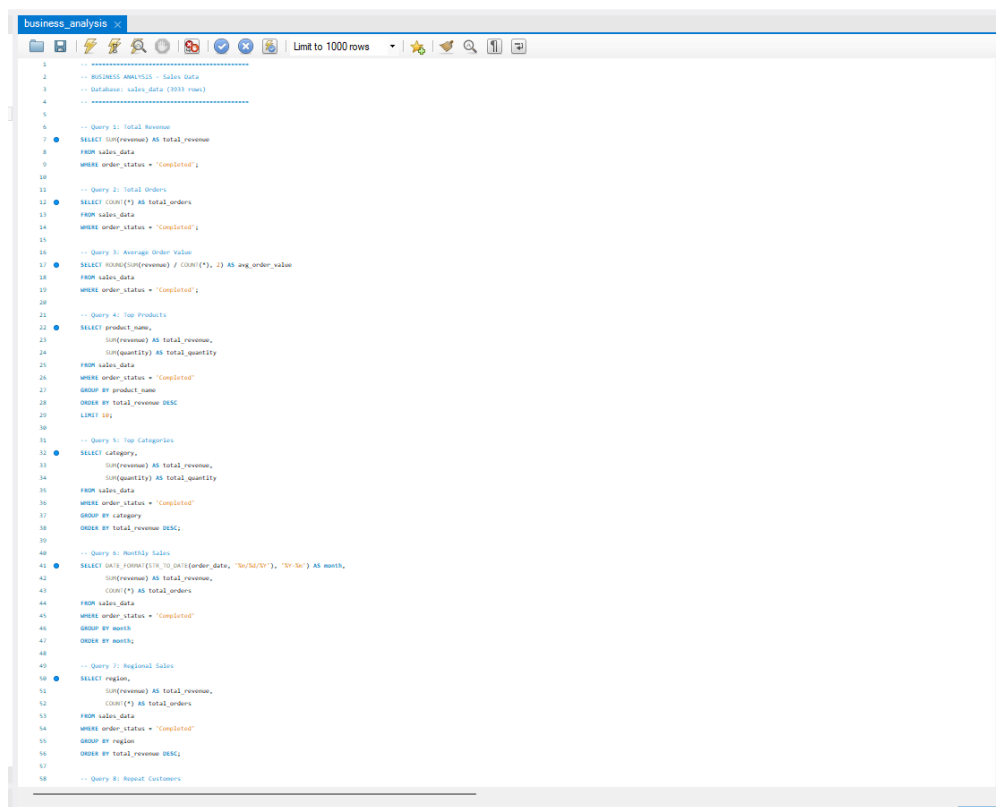
```

month	total_revenue	cumulative_revenue
2023-01	10999.53	10999.53
2023-02	11485.76	22485.29
2023-03	13600.22	36085.51
2023-04	10068.15	46153.66
2023-05	12690.06	58843.72
2023-06	7248.67	66092.39
2023-07	11168.56	77260.95
2023-08	11470.69	88731.64
2023-09	12740.75	101472.39
2023-10	11436.65	112909.04
2023-11	9873.17	122782.21
2023-12	7075.01	129857.22
2024-01	9416.06	139273.28
2024-02	8862.26	148135.54
2024-03	10659.45	158794.99
2024-04	9007.92	167802.91
2024-05	12166.04	179968.95
2024-06	8924.90	188893.85
2024-07	9983.49	198877.34
2024-08	10466.35	209343.69
2024-09	8546.15	217889.84
2024-10	11485.51	229375.35
2024-11	11660.19	241035.54
2024-12	11192.17	252227.71

Gambar 18. Hasil Query 15 — cumulative_revenue akhir (Des 2024) = 252.227,71, konsisten dengan Query 1

3.18 Menyimpan Query — business_analysis.sql

Seluruh 15 query digabung dalam satu file dan disimpan lewat File → Save Script As di MySQL Workbench, agar mudah dipakai ulang atau di-review orang lain.



```

1  -- =====
2  -- BUSINESS ANALYSIS - Sales Data
3  -- Database: sales_data (3933 rows)
4  -- =====
5
6  -- Query 1: Total Revenue
7  SELECT SUM(revenue) AS total_revenue
8  FROM sales_data
9  WHERE order_status = 'Completed';
10
11 -- Query 2: Total Orders
12 SELECT COUNT(*) AS total_orders
13 FROM sales_data
14 WHERE order_status = 'Completed';
15
16 -- Query 3: Average Order Value
17 SELECT ROUND(SUM(revenue) / COUNT(*), 2) AS avg_order_value
18 FROM sales_data
19 WHERE order_status = 'Completed';
20
21 -- Query 4: Top Products
22 SELECT product_name,
23        SUM(revenue) AS total_revenue,
24        SUM(quantity) AS total_quantity
25 FROM sales_data
26 WHERE order_status = 'Completed'
27 GROUP BY product_name
28 ORDER BY total_revenue DESC
29 LIMIT 10;
30
31 -- Query 5: Top Categories
32 SELECT category,
33        SUM(revenue) AS total_revenue,
34        SUM(quantity) AS total_quantity
35 FROM sales_data
36 WHERE order_status = 'Completed'
37 GROUP BY category
38 ORDER BY total_revenue DESC;
39
40 -- Query 6: Monthly Sales
41 SELECT DATE_FORMAT(ORDER_DATE, '%Y-%m') AS month,
42        SUM(revenue) AS total_revenue,
43        COUNT(*) AS total_orders
44 FROM sales_data
45 WHERE order_status = 'Completed'
46 GROUP BY month
47 ORDER BY month;
48
49 -- Query 7: Regional Sales
50 SELECT region,
51        SUM(revenue) AS total_revenue,
52        COUNT(*) AS total_orders
53 FROM sales_data
54 WHERE order_status = 'Completed'
55 GROUP BY region
56 ORDER BY total_revenue DESC;
57
58 -- Query 8: Repeat Customers

```

Gambar 19. File `business_analysis.sql` tersimpan lengkap dengan komentar per query

4. Business Insights

Delapan insight berikut disusun berdasarkan seluruh angka dan pola yang ditemukan sepanjang analisis SQL di atas.

1. Ketergantungan Ekstrem pada Region East

Region East menyumbang 74,4% dari total revenue (187.615,92 dari 252.227,71) dan 74,3% dari total order (2.394 dari 3.220), sementara West cuma menyumbang 1,1% (2.855,68, 32 order). Ketimpangan seekstrem ini menunjukkan bisnis sangat bergantung pada satu wilayah — risiko besar kalau performa East menurun.

2. Nilai Transaksi Tidak Selalu Sejalan dengan Frekuensi Order

Customer paling sering order (Ayu Wilson, 11x) bukan customer dengan belanja terbesar. Andi Salim justru mencatat total spending tertinggi (1.285,26) hanya dari 6 order — menandakan ada segmen customer bernilai transaksi besar yang berbeda dari segmen customer paling loyal.

3. Electronics Adalah Kategori dan Produk Dominan

Electronics adalah kategori dengan revenue tertinggi (63.888,07, 25,33% dari total), diikuti Home & Kitchen (51.020,37, 20,23%). Produk terlaris (Smartphone Case, Webcam, Power Bank) juga berasal dari kategori Electronics — kategori ini jadi pendorong utama penjualan.

4. Books Jadi Kategori dengan Performa Terlemah

Books adalah kategori dengan revenue terendah (15.737,68, 6,24% dari total), kurang dari seperempat revenue Electronics. Produk-produk seperti Business Strategy Book dan Fiction Novel juga menempati peringkat revenue terendah secara individual — kategori ini berpotensi dievaluasi ulang dari sisi variasi produk atau strategi promosi.

5. Tidak Ada Pola Musiman yang Konsisten

Growth month-over-month berfluktuasi tajam tanpa pola musiman yang jelas — dari -42,88% (Juni 2023) sampai +54,08% (Juli 2023). Ketidakstabilan ini menunjukkan revenue lebih dipengaruhi faktor acak/operasional dibanding tren musiman yang bisa diprediksi.

6. Revenue Kumulatif Tumbuh Stabil Meski Fluktuatif Bulanan

Meski revenue bulanan naik-turun tajam, tren cumulative revenue (Query 15) menunjukkan pertumbuhan yang stabil dan konsisten sepanjang 24 bulan, mengonfirmasi bisnis tetap bertumbuh secara agregat walau performa bulan-ke-bulan tidak merata.

7. Penjualan Merata Sepanjang Minggu

Tidak ada hari yang mendominasi secara ekstrem — revenue tertinggi (Senin, 38.021,54) dan terendah (Sabtu, 34.456,04) hanya berselisih sekitar 10%. Ini menunjukkan pola belanja customer tidak terlalu dipengaruhi hari dalam seminggu.

8. AOV Tergolong Rendah, Peluang Bundling/Cross-Sell

Average Order Value hanya 78,33 dari 3.220 order Completed, jauh lebih rendah dibanding average revenue per customer (288,59) — artinya banyak customer kembali berkali-kali dengan nilai transaksi kecil per kunjungan. Ada peluang menaikkan AOV lewat strategi bundling atau cross-selling produk pelengkap.

5. Ringkasan Checklist Day 3

Berikut status penyelesaian setiap tahapan pada Day 3, diambil dari tracker progress portofolio.

No	Hari	Tahap	Task	Tools	Status
28	Day 3	SQL	Import dataset ke MySQL	MySQL	SELESAI
29	Day 3	SQL	Query: Total revenue	MySQL	SELESAI
30	Day 3	SQL	Query: Total orders	MySQL	SELESAI
31	Day 3	SQL	Query: Average order value	MySQL	SELESAI
32	Day 3	SQL	Query: Top products	MySQL	SELESAI
33	Day 3	SQL	Query: Top categories	MySQL	SELESAI
34	Day 3	SQL	Query: Monthly sales	MySQL	SELESAI
35	Day 3	SQL	Query: Regional sales	MySQL	SELESAI
36	Day 3	SQL	Query: Repeat customers	MySQL	SELESAI
37	Day 3	SQL	Query: Highest-value customers	MySQL	SELESAI
38	Day 3	SQL	Query: Lowest-performing products	MySQL	SELESAI
39	Day 3	SQL	Query: Growth MoM	MySQL	SELESAI
40	Day 3	SQL	Query: Category contribution	MySQL	SELESAI
41	Day 3	SQL	Query: Average revenue per customer	MySQL	SELESAI
42	Day 3	SQL	Query: Sales by weekday	MySQL	SELESAI
43	Day 3	SQL	Query: Sales trend	MySQL	SELESAI
44	Day 3	SQL	Total minimal 15 query terkumpul	MySQL	SELESAI
45	Day 3	SQL	Simpan sebagai business_analysis.sql	MySQL	SELESAI

Tabel 2. Status seluruh task Day 3 SQL Analysis

6. Kesimpulan

Analisis Day 3 menghasilkan business_analysis.sql berisi 15 query bisnis terdokumentasi — mulai dari metrik dasar (total revenue, total order, AOV) sampai analisis lanjutan menggunakan window function (growth month-over-month, cumulative revenue trend). Seluruh proses dibangun di atas dataset bersih 3.933 baris dari Day 1, dan seluruh angka hasil query saling konsisten satu sama lain (misalnya total_revenue Query 1 sama persis dengan cumulative_revenue akhir di Query 15).

Temuan paling signifikan: ketergantungan ekstrem pada region East (74,4% revenue), kategori Electronics sebagai pendorong utama penjualan, dan AOV yang tergolong rendah dibanding average revenue per customer — tiga poin ini melengkapi dan memperkuat insight yang sudah ditemukan di Day 2, serta menjadi kandidat kuat untuk dibahas lebih dalam di Day 6 (Business Case Study).

File business_analysis.sql ini menjadi pembanding/validasi silang untuk Day 4 (Python EDA) dan Day 5 (Tableau dashboard) — khususnya angka-angka KPI dan insight di atas akan dipakai untuk memastikan konsistensi hasil analisis di setiap tahap.